

Advanced Python — Unit 2

NumPy • Hard 2-Mark Questions

Instruction: Predict the output or answer as specified for each question.

Q1. [2 Marks]

```
import numpy as np

a = np.array([[2, 4, 6],
              [1, 3, 5],
              [8, 10, 12]])

print(a[1:, :2])
```

Q2. [2 Marks]

```
import numpy as np

a = np.array([[2, 4, 6],
              [1, 3, 5],
              [8, 10, 12],
              [7, 9, 11]])

b = np.array([1, 2, 3])

x = a[:,2, 1:]

print(x)
print(x * b[1:])
```

Q3. [2 Marks]

```
import numpy as np

a = np.array([[1, 2, 3],
              [4, 5, 6]])

b = np.array([[10],
              [20]])

print(a + b)
```

Q4. [2 Marks]

```
import numpy as np

a = np.array([[2, 4, 6],
              [8, 10, 12]])

print(a[a > 5])
```

Q5. [2 Marks]

```
import numpy as np

a = np.array([[1, 2, 3],
              [4, 5, 6]])

print(a.sum(axis=0))
print(a.sum(axis=1))
```

Q6. [2 Marks]

```
import numpy as np

a = np.array([[1, 2],
              [3, 4]])

b = np.array([10, 20])

print(a + b)
```

Q7. [2 Marks]

```
import numpy as np

a = np.arange(1, 13).reshape(3, 4)

x = a[:, [1, 3]]

print(x)
print(x.shape)
```

Q8. [2 Marks]

```
import numpy as np

a = np.array([2, 5, 8, 11, 14])

x = a[(a > 5) & (a < 12)]

print(x)
```

Q9. [2 Marks]

```
import numpy as np

a = np.array([[1, 2, 3],
              [4, 5, 6]])

x = a.T

print(x[1:, :])
```

Q10. [2 Marks]

```
import numpy as np

a = np.array([[1, 2, 3],
              [4, 5, 6]])

b = np.array([1, 2, 3])

x = a[:, ::2]

print(x)
print(x + b[::2])
```